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Enabling Cybersecurity Incident Response Agility Through Dynamic Capabilities: The Role of Real-Time Analytics

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Enabling Cybersecurity Incident Response Agility Through Dynamic Capabilities: The Role of Real-Time Analytics

We explore how organisations enable agility in their cybersecurity incident response (IR) process by developing dynamic capabilities using real-time analytics (RTA). Drawing on RTA practices in the IR process at three large financial organisations, we develop a framework to explain how IR teams respond to the rapidly evolving cyber threat environment by developing RTA-based microfoundations that underpin the building of sensing, seizing, and transforming dynamic IR capabilities. These dynamic IR capabilities in turn help organisations to enable agility in their IR processes by leveraging swift, flexible, and innovative IR strategies, including *active threat reconnaissance*, *active threat defence*, and *pervasive learning*. Our findings have implications for the discourse on cybersecurity because we demystify the black box of IR agility, for our understanding of the use of RTA to enable agility in IR, and for the discourse on dynamic capabilities.

Keywords: Real-time analytics, Cybersecurity, Incident response, Agility, Dynamic capabilities

1. Introduction

We have to keep on remembering that the attacks are going to come. The attackers are continuously innovating, and we must do the same. We have to be agile in our response, and that requires sharing the right information with the right people at the right time to be able to respond in the right manner. (FinBank, Chief Information Security Officer)

The prevailing fast-paced and highly dynamic business environment shapes how organisations use their enterprise assets and capabilities such as digital processes, technologies, data, and information to gain a competitive advantage (Hagiu & Wright, 2020). While organisations perform their daily business operations, these assets are increasingly exposed to complex and evolving cybersecurity threats, both internal and external, such as theft, fraud, embezzlement, sabotage, and industrial espionage (McMillan & Proctor, 2018). A successful cybersecurity

attack on enterprise assets can result in major damage to organisations in terms of financial loss (from theft of information or money, and disruption to business), reputational damage (from loss of customer trust), and legal consequences (Goode et al., 2017; Verizon, 2022). It is, therefore, critical for organisations to detect and respond to cybersecurity incidents before they can cause any damage.

Organisations employ cybersecurity incident response (IR) teams to detect, contain, eradicate, and recover from cybersecurity incidents (Kotsias et al., 2022). The overarching IR process consists of a collection of procedures aimed at identifying, investigating, and responding to potential cybersecurity threats in a manner that minimises impact and supports rapid recovery (Creasy & Glover, 2013). In this research, we argue that the dynamic and evolving nature of contemporary cyber threats demands a higher degree of cybersecurity awareness and a sophisticated, timely, and agile response capability to counter the evolving vectors of the attack (Baskerville et al., 2014; Yeoh et al., 2022). This requires an organisation's IR function to have processes, tools, and skills that enable the enterprise-wide collection, integration, and analysis of information relevant to cybersecurity threats to inform timely strategic, tactical, and operational cybersecurity decisions (H. Naseer et al., 2021).

Business analytics (BA) is an organisational capability integrating people, processes, and technology to generate insights, empowering business executives to make informed and timely decisions (Conboy et al., 2020; Seddon et al., 2017; Wixom et al., 2013). BA capability is suited to collecting, integrating, and analysing large sets of disparate organisational data (Grover et al., 2018; Tim et al., 2020). A specialised BA capability—real-time analytics (RTA)—focuses on analysing streaming data, thereby enabling organisations to carry out BA in real-time and proactively make informed business decisions (Müller et al., 2016; A. Naseer et al., 2021; Phillips-Wren et al., 2015). We argue that RTA provides organisations with a unique opportunity to detect and analyse ongoing cybersecurity incidents as they happen and respond to them in a timely and proactive manner.

Traditionally, most organisations use RTA to enhance their operational competencies (Townsend et al., 2018). However, some enterprises have recognised that BA capabilities enable a fundamental shift in the way that information technology (IT) capabilities are delivered and can be used to enable agility by developing dynamic capabilities (Conboy et al., 2020; Park et al., 2017). Dynamic capabilities are innovation-based and provide the ability to create, modify, and extend a firm's resource base in rapidly changing environments (Pavlou & El Sawy, 2011; Teece et al., 2016). Given the uncertain, unknown, and evolving nature of the cyber threat environment, we posit that the dynamic capabilities framework is a powerful lens for examining how agility manifests in the IR process using RTA. In particular, it has been reported that organisations need to develop dynamic capabilities in cybersecurity environments that face dynamic, unknown, and evolving attacks (Baskerville et al., 2014; H. Naseer et al., 2018). However, in the context of the IR process, *how* organisations build dynamic capabilities using RTA, and thereby enable agility in the IR process, is not yet well explored (A. Naseer et al., 2021).

Therefore, our research aims to demystify the black box of enabling agility in IR. Without a clear understanding of how agility is enabled in IR, organisations may struggle to manage cybersecurity incidents swiftly, leading to prolonged downtime, financial losses, and reputational damage (Verizon, 2022). Accordingly, we ask: *How do organisations enable cybersecurity incident response agility by developing dynamic capabilities using real-time analytics?* To answer this, data from three large financial organisations that use RTA in their cybersecurity IR processes were analysed. Integrating insights from the data with the existing literature informed a framework that identifies nine RTA-based microfoundations that underpin the development of sensing, seizing, and transforming dynamic IR capabilities. The details of the framework advance our understanding of what dynamic IR capabilities are required to enable agility in the IR process.

In the following sections, we first present the theoretical background of our work. Next, we introduce the research design of our study, and then present our findings. Finally, we discuss the implications and limitations of our work and provide a brief conclusion.

2. Theoretical background

In this section, we summarise the literature on cybersecurity IR, including the relatively few studies that examine the role of BA in the IR process. Then, we introduce dynamic capabilities as a theoretical lens and explain how it can be adapted to the cybersecurity IR context and how it allows us to conceptualise IR agility.

2.1. The cybersecurity incident response process

Cybersecurity refers to the practice of protecting enterprise assets and digital information from unauthorised access, disclosure, misuse, disruption, modification, or destruction (von Solms & van Niekerk, 2013; Yeoh et al., 2022). Organisations use a combination of technologies, strategies, and user education to protect their assets against cybersecurity attacks that can steal valuable information, compromise systems, and damage reputation (Jalali et al., 2019; Khan et al., 2021; Wong et al., 2020). According to Lemay et al. (2018), as the severity and volume of cyber-attacks increase, the need for cybersecurity increases with them.

To counter cybersecurity attacks and data breaches, organisations engage their IR teams (ad-hoc or dedicated) to detect and eradicate threats (Creasy & Glover, 2013; Kotsias et al., 2022). A cybersecurity incident is an unexpected or unwanted cybersecurity event or a series of events that have a significant probability of compromising business operations (Cichonski et al., 2012; Mitropoulos et al., 2006). To ensure the continuity of business operations, IR is the immediate action that IR teams execute to deal with cyber-attacks (Ruefle et al., 2014; Siregar & Chang, 2019). Therefore, swift and agile execution of the cybersecurity IR process is crucial and requires a variety of qualities in IR teams such as technical, analytical, and communication skills (A. Naseer et al., 2021). A security operations centre (SOC) is usually established by

organisations, from where IR teams monitor, detect, analyse, and respond to potential cybersecurity incidents, typically on a round-the-clock basis (Cichonski et al., 2012; Creasy & Glover, 2013; Kotsias et al., 2022).

2.2. Real-time analytics in cybersecurity incident response

Organisations are increasingly adopting BA practices in the IR process to make informed cybersecurity-related decisions (Pierazzi et al., 2016; Wang & Jones, 2021). These practices enable three types of analytics in IR (Davenport et al., 2010; H. Naseer, Maynard, et al., 2016): 1) descriptive analytics, which uses historical data to provide trending information about past or current cybersecurity events; 2) predictive analytics, which provides likely scenarios to cybersecurity managers so that they can plan ahead, rather than react to what has already happened; and 3) prescriptive analytics, which explores a set of possible actions and suggests a reliable path to an optimal solution based on descriptive and predictive analyses of complex data (Eastman & Versace, 2015; H. Naseer et al., 2021).

Descriptive, predictive, and prescriptive analytics provide cybersecurity insights for both historical analysis and future planning and are, therefore, valuable to organisations. However, these insights have latencies (such as data, analysis, and decision latencies), because data first need to be stored and then processed to generate insights (Phillips-Wren et al., 2015). Therefore, these insights are not ideal for cybersecurity events that need to be analysed as they occur (e.g., zero-day attacks). To analyse and monitor business events as they occur, BA scholars have proposed the specialised BA capability known as RTA (Grover et al., 2018; Wixom et al., 2013).

The concept of RTA has existed in the cybersecurity literature and practice for a long time (Mahmood & Afzal, 2013); however, understanding of this concept is still limited in terms of defining the meaning of “real-time” in the IR context and what additional components are required in typical security analytics architecture to enable RTA in the IR process (A. Naseer et al., 2021). In addition, for organisations to leverage RTA to strengthen their IR processes, they must begin to develop new capabilities (using RTA) that enable swift detection and eradication

of cybersecurity threats. Therefore, the literature from strategic management related to firm-level resources and the development of dynamic capabilities provides a sound basis for studying how organisations use RTA to develop dynamic capabilities that are necessary to achieve agility in IR.

2.3. Dynamic capabilities in cybersecurity incident response

Dynamic capabilities are an organisation's "ability to integrate, build and reconfigure internal and external competences to address rapidly changing environments" (Teece et al., 1997, p. 516).

Dynamic capabilities theory was originally developed to overcome the limitations set by the static orientation of the resource-based view (RBV); however, it is now widely regarded as an extension of RBV (Schilke & Helfat, 2018; Steininger et al., 2022). While RBV argues that organisations can achieve a competitive advantage based on their bundles of resources and capabilities, dynamic capabilities theory proposes that organisations must evolve their resource and capability base to ensure a sustained competitive advantage (Schilke & Helfat, 2018).

Dynamic capabilities are innovation-based and purposefully built and developed rather than acquired and applied (Helfat & Winter, 2011; Teece et al., 2016). The concept of microfoundations within the dynamic capabilities framework provides an explanation for the origins and development of dynamic capabilities (Barney & Felin, 2013). According to Eisenhardt et al. (2010, p. 1263), microfoundations are the underlying actions, activities, and interactions at individual and group levels that shape strategy and organisation as well as dynamic capabilities, leading to the emergence of superior organisation-level performance. Dynamic capabilities can be bundled into three broad clusters of microfoundations: (1) *sensing* opportunities and threats; (2) *seizing* opportunities; and (3) *transforming* or reconfiguring an organisation's resources and capabilities (Teece, 2007).

The dynamic capabilities lens is adaptable to the cybersecurity IR context via manipulation of the focal outcome variable. While the majority of the dynamic capabilities

literature examines its role in generating new value-creating strategies and/or competitive advantage (Schilke & Helfat, 2018; Steininger et al., 2022), cybersecurity IR aims to protect business value/competitive advantage (Goel et al., 2023; Kosutic & Pigni, 2022; H. Naseer, Shanks, et al., 2016). The role of dynamic capabilities in protecting business value/competitive advantage has been discussed in other research areas such as enterprise risk management, disaster and crisis management, and organisational cyber resilience (Battisti & Deakins, 2017; Ferdinand, 2015; Goel et al., 2023; Jiang et al., 2021). Based on the three dynamic capabilities of microfoundations (sensing, seizing, and transforming), Jiang et al. (2021) propose a dynamic resilience framework to explain how tourism organisations develop disaster resilience elements at each disaster management stage. Goel et al. (2023) conceptualise information systems security resilience as a dynamic capability of a firm to respond to, and recover from, a security attack and test it as a new construct. Battisti and Deakins (2017) provide empirical evidence that small firms need to make proactive efforts to integrate resources from external networks for the dynamic capability to survive and recover from a disaster situation. Ferdinand (2015) uses the dynamic capabilities framework to explain how an organisation can build, assess, and maintain cyber resilience to sustain competitive advantage. H. Naseer et al. (2018) propose a dynamic cybersecurity risk management framework to explain how organisations improve their enterprise security performance by developing dynamic capabilities in cybersecurity risk management. In all the aforementioned studies, the central role of dynamic capabilities is to enable the continuity of business operations through the protection of organisational assets such as infrastructure, human resources, data, and information. Further, via this protection, the dynamic capabilities represent a differentiating factor for organisations in the event of a disaster or cyber-attack. Therefore, we suggest that the central role of dynamic capabilities in cybersecurity IR is to enable the continuity of business operations by protecting critical enterprise systems and data assets against malicious cyber threats, thereby safeguarding organisational performance.

While prior research has primarily explored the enablers of dynamic capabilities (Karimi-Alagheband & Rivard, 2019), numerous studies have also examined the outcomes of dynamic capabilities (Battleson et al., 2016; Jantunen et al., 2018). These studies have considered antecedents at different levels of analysis, including at the individual, organisational, and environmental levels, to identify factors that either facilitate or hinder the formation of dynamic capabilities (Schilke & Helfat, 2018; Steininger et al., 2022). However, to the best of our knowledge, no research has examined how a specialised BA capability (in this case RTA) helps organisations to build the microfoundations of sensing, seizing, and transforming dynamic capabilities in the process of cybersecurity IR.

2.4. Conceptualising incident response agility through dynamic capabilities

The IR process is a nuanced activity that involves high levels of complexity and uncertainty created by the evolving nature of cybersecurity threats (Desouza et al., 2020; Tøndel et al., 2014). Our review of the IR literature highlights that the focus of IR strategies in many organisations is to invest in sophisticated preventative controls aimed at combating known threats rather than developing agile response processes to address unknown, complex, and evolving threats (Baskerville et al., 2014; Shin & Lowry, 2020). The commercial deliberations also suggest a lack of agility in executing IR processes in a real-world cybersecurity incident-handling context (Creasy & Glover, 2013; Grispos et al., 2014; He et al., 2022; McMillan & Proctor, 2018). IR agility is defined as “the extent to which an organisation can add and/or reconfigure its IR resources and processes to detect and respond to unpredictable, unknown and new cybersecurity threats” (A. Naseer et al., 2021, p. 7). IR agility emphasises the need for an enterprise to detect malicious cyber threats quickly and then provide focused and swift responses to these threats by reconfiguring the IR processes and resources.

Although organisations are increasingly paying attention to enabling IR agility, how agility can be achieved in the IR process is still a black box (Baskerville et al., 2014; A. Naseer et

al., 2021). Teece et al. (2016) highlight that agility is crucial to managing deep uncertainty or an unknown–unknown event and offer a dynamic capabilities framework to explain how agility is manifested in the organisational context. A cybersecurity incident is an instance of deep uncertainty and unknown–unknown cybersecurity events can have a significant probability of compromising business operations (Jalali et al., 2019; Verizon, 2022). Thus, we argue that organisations need agility in their IR process to deal with cybersecurity incidents proactively. Organisational agility is manifested by, and focuses on, supporting the enterprise-wide strategic activities of sensing and responding to external and internal business events of environmental change in a timely manner to seize opportunities and handle threats efficiently and effectively (Overby et al., 2006).

The existing literature suggests that BA plays a central role in enabling dynamic capabilities and achieving organisational agility (Conboy et al., 2020; Park et al., 2017). Extending the aforementioned observations on BA as an enabler of dynamic capabilities and organisational agility to a specialised BA capability (in this case, RTA) and a specific form of organisational agility (in this case, business process agility), this study seeks to investigate how agility is manifested in the cybersecurity IR process through the development of dynamic capabilities using RTA.

3. Research methodology

3.1. Research design and context

We chose a case study design to explore how organisations enable agility in cybersecurity IR by developing dynamic capabilities using RTA (Eisenhardt, 1989; Yin, 2017). Theory building through case studies is an appropriate research strategy when little *a priori* knowledge exists about a phenomenon (Benbasat et al., 1987; Yin, 2017). We specifically employed a multiple case study design because this enabled us to compare cases and clarify whether an emergent finding was consistent across cases (Eisenhardt & Graebner, 2007).

3.2. Case companies' background and selection

We sought organisations that were using BA to address their evolving cyber threat environment. The financial sector presented as an exemplary industry. Reports and discussions with cybersecurity experts suggested that financial organisations face a rapidly evolving cyber threat environment and, therefore, develop analytical capabilities to deal with such an environment. In addition, financial sector organisations spend the most revenue on cybersecurity and, thereby, are at high levels of maturity in their BA practices in the IR process (AustCyber, 2022). We employed theoretical sampling to select organisations with sophisticated RTA and cybersecurity solutions, such as security information and event management (SIEM), complex event processing, real-time analysis, and reporting tools (Glaser & Strauss, 2017). The goal of theoretical sampling was to choose cases that were likely to replicate or extend the emergent theory (Eisenhardt, 1989; Yin, 2017). We refer to the case organisations as FinBank, FinInsuranceA, and FinInsuranceB to protect the identities of the participants. Table 1 provides the business context and background story of each firm.

Table 1. The profile, business context, and background story of the case organisations

ID: FinBank	Organisation type: Bank	Number of employees in SOC: 25
FinBank is a large financial company in Australia and a major international banking and financial services group within the top 50 banks worldwide. At FinBank, BA practices are integrated into the IR function to nurture a data-driven IR decision-making culture. Finbank's IR function manages approximately 15 billion cybersecurity events each day that are generated by its financial services technology ecosystem. FinBank's cybersecurity strategy includes a SOC, SIEM systems, and a custom-built security analytics solution helping to generate insights based on security data and taking informed actions. FinBank uses BA in each phase of its cybersecurity IR process. Examples of where FinBank's cybersecurity IR team uses BA include: (a) access management (e.g., in blocking and logging of unauthorised access); (b) forensic analysis (e.g., analysing network data to monitor network activity in real-time); and (c) analysis of logs (e.g., service and application logs, operating system logs, network device logs, and network flows).		
ID: FinInsuranceA	Organisation Type: Insurance	Number of Employees in SOC: 18
FinInsuranceA is an Australian-based large multinational insurance company. Since 2015, FinInsuranceA has invested heavily in building a data-driven decision-making capability in its cybersecurity IR function. FinInsuranceA has a dedicated analytics team that has built its security analytics capability. The FinInsuranceA IR function manages approximately 10 billion cybersecurity events each day. FinInsuranceA's cybersecurity strategy includes a SOC, SIEM, and managed security service provider. FinInsuranceA also uses BA in all phases of the cybersecurity IR process. Examples of where FinInsuranceA's cybersecurity IR team uses BA include: (a) security log analysis to correlate security events; (b) access management (e.g., managing access across all its systems and assets in real-time); and (c) malware analysis.		
ID: FinInsuranceB	Organisation Type: Insurance	Number of Employees in SOC: 20
FinInsuranceB is a United States-based multinational financial services company and has financial services branches		

situated worldwide. FinInsuranceB has a cybersecurity division that supports multiple business domains with a dedicated SOC and analytics team, collaboratively working with business functions. The cybersecurity analytics division was established in 2016 to address insider-threat and data-leakage issues in the organisation. The FinInsuranceB IR function manages approximately 12 billion cybersecurity events each day. FinInsuranceB's cybersecurity strategy includes a combination of SOC and SIEM systems. FinInsuranceB has developed an advanced cyber analytics capability that delivers insights to security executives for informed decision-making related to cybersecurity. FinInsuranceB also uses BA in all phases of the cybersecurity IR process. Examples of where FinInsuranceB's cybersecurity IR team uses BA include: (a) insider threat detection; (b) employee activity monitoring; and (c) analysing and monitoring emerging cyber threats.

3.3. Data collection

Face-to-face semi-structured interviews were conducted at each organisation to collect data (Alvesson, 2003). Relevant background information, such as organisational documents, strategic progress reports, internal publications, and PowerPoint presentations, was made available and collected as complementary material. In addition, each organisation's public presence was investigated, and data were gathered from their websites and industry analysis reports to enable triangulation around the understanding of each organisation's IR function strategy, practices, processes, and outcomes. To triangulate insights from different sources (Eisenhardt & Graebner, 2007) and mitigate informant biases (Miller et al., 1997), we interviewed participants at various levels in the cybersecurity unit, including strategic top-level executives, middle-level senior managers, and cybersecurity analysts working with the IR function. In total, 27 interviews were conducted.

We first interviewed senior executives and discussed the research objectives with them. They then helped to identify additional participants who were selected for their extensive experience in both the analytics and cybersecurity domains, with appropriate knowledge to provide insights on behalf of their organisation (see Table 2). Appendix A provides examples of open-ended questions that guided our data collection. Interviews were recorded and transcribed (with participant consent) and lasted between one and two hours. As the data collection process continued, we continuously revised the interview protocol based on informant responses (Gioia et al., 2013). As the interview concluded, participants were asked for further reflections on the examples they gave during the interview. During the interviews, we also took detailed notes and

photographs¹ to supplement the data. In addition, onsite observations (recorded using detailed field notes) of how the SOC was set up—including what the different tiers were, where they were located and how they interacted—provided further complementary insights into the context of our study (Klein & Myers, 1999).

Table 2. Summary of informant details

Firm		Description of interviewees	Experience (BA, cybersecurity)
FinBank 11 Interviewees	Top-level executives	Chief Information Security Officer	10 yrs, 20 yrs
		General Manager of Cybersecurity Risk Management	25 yrs, 14 yrs
		General Manager of Data and Analytics	20 yrs, 10 yrs
	Middle-level managers	Cybersecurity Senior Manager	8 yrs, 12 yrs
		Cybersecurity Senior Manager	7 yrs, 14 yrs
		Head of Cybersecurity Incident Response Team	6 yrs, 10 yrs
		Security Operations Centre Manager	6 yrs, 11 yrs
	Cybersecurity analysts	Senior Cybersecurity Analyst	5 yrs, 9 yrs
		Senior Cybersecurity Analyst	6 yrs, 8 yrs
		Cybersecurity Analyst	2 yrs, 4 yrs
		Cybersecurity Analyst	1 yr, 2 yrs
FinInsuranceA 9 Interviewees	Top-level executives	General Manager of Cybersecurity Strategy and Governance	12 yrs, 20 yrs
		General Manager of IT and Information Security	10 yrs, 18 yrs
	Middle-level managers	Manager of Cybersecurity Strategy and Governance	8 yrs, 15 yrs
		Manager of Cyber Defence Centre	8 yrs, 12 yrs
		Head of Cyber Threat Detection and Response Team	6 yrs, 13 yrs
		Senior IT Risk Partner	2 yrs, 7 yrs
		Senior IT Risk Partner	3 yrs, 6 yrs
	Cybersecurity analysts	Senior Cybersecurity Analyst	2 yrs, 6 yrs
		Cybersecurity Analyst	1 yr, 2 yrs
FinInsuranceB 7 Interviewees	Top-level executives	Director of Global Cyber Forensics	15 yrs, 13 yrs
		Chief Security Architect	10 yrs, 18 yrs
	Middle-level managers	Head of Global Insider Threat Detection	6 yrs, 8 yrs
		Head of Cybersecurity Incident Response Team	5 yrs, 9 yrs
		Cybersecurity Senior Manager	6 yrs, 10 yrs
	Cybersecurity analysts	Cybersecurity Analyst	2 yrs, 2 yrs
		Cybersecurity Analyst	1 yr, 2 yrs

Note: Total number of interviews = 27

¹ Some of the informants used additional tools such as whiteboard/marker, pen/paper, and posters already displayed in the interview space to further elaborate their responses. We photographed this supplementary material after seeking consent.

3.4. Data analysis

The data analysis and data collection processes were conducted concurrently to allow for the emergence of empirical data and theoretical concepts in capturing a novel phenomenon. While the interviews formed the primary source of data analysis, they were corroborated by onsite observations, field notes, and other available data, such as strategic progress reports, organisational documents, PowerPoint presentations, and internal publications. We analysed our data following the Gioia methodology, which draws on the principles of grounded theory (Gioia et al., 2013; Strauss & Corbin, 2014). The analysis was carried out through a three-stage process.

In the first stage, we employed an “induction” method and coded to develop first-order concepts by adhering to informant-centric words and terms (Gioia et al., 2013; Kennedy & Thornberg, 2018). At this stage, we let patterns and concepts emerge from the data without presupposing outcomes *a priori*. For example, the codes captured concepts such as “real-time cybersecurity awareness”, “continuous event monitoring”, “uncertain cyber threat environment”, “real-time cyber threat intelligence”, “proactive risk assessment” and “dynamic incident response”. We assigned these codes to words, sentences, or paragraphs in the interview transcripts. The data-coding process and interviewing continued until theoretical saturation was reached (i.e., no further distinct, shared patterns were found in the data) (Gioia et al., 2013). Overall, 250 codes were assigned to explain how RTA was used by IR teams in practice to address the changing cyber threat environment.

In our initial data analysis, we observed that IR teams deemed IR agility as the key attribute in addressing the uncertain and evolving cyber threat environment. This observation directed us to consider the role of dynamic capabilities in enabling agility in the IR process. Therefore, in the second stage, we transitioned from induction to “abduction” as the logical data analysis process and used the existing literature and our knowledge to analyse and explain the relationships between emergent concepts (Gioia et al., 2013; Kennedy & Thornberg, 2018). At

this stage, we used dynamic capabilities as a theoretical frame of reference and sensitising device to develop theoretically informed second-order themes. We reduced the first-order concepts by deriving second-order themes representing strategic activities related to RTA and microfoundations sensitised by the overarching dimensions of the dynamic capabilities framework. While we did not allow prior theoretical concepts and assumptions to restrict our interpretations, they helped us to structure and make sense of the data. For example, the first-order concepts “real-time cybersecurity awareness”, “continuous event monitoring”, and “real-time cyber threat intelligence” described activities and actions that IR teams took at both individual and group levels to analyse diverse information about cyber threats in real-time. Therefore, we developed “situation awareness”, “threat intelligence”, and “continuous monitoring” as the second-order themes to capture the microfoundations for *sensing* dynamic capabilities enabled by RTA that helped in scanning, learning, and interpreting cyber threats in the evolving cyber threat environment. By doing so, we combined similar first-order concepts across cases into a set of second-order themes that represented the microfoundations of sensing, seizing, and transforming dynamic capabilities in the IR process. We triangulated the second-order themes and first-order concepts with supplementary interviews to strengthen the contextuality of our theoretical framework (Gioia et al., 2013). Through the iterative interplay between (a) data collection and analysis, in which ongoing analysis of data guided our further investigation, and (b) data and literature, in which we used dynamic capabilities as the sensitising device to describe and explain the phenomenon we were observing, we were able to rigorously develop detailed and accurate descriptions of concepts from empirical data (Alvesson & Kärreman, 2007; Kennedy & Thornberg, 2018).

Finally, we further analysed the data, aggregating the concepts identified in second-order themes to form higher-level abstract concepts (aggregate dimensions). This allowed us to create a data structure that represents our analysis graphically, from the raw data to aggregated dimensions, and configures the data into a sensible visual aid (Gioia et al., 2013). The first-order

concepts incorporate the language of the participants and describe the key activities to enable agility in the IR process. The final data structure (see Figure 1) summarises the interrelations of first-order concepts, second-order themes, and aggregate dimensions. From these, we propose a theoretical framework (see Figure 2) that captures the informants' experience in theoretical terms and shows dynamic relationships among the emergent concepts. Using the data structure, we revalidated the final concepts back to the underlying data and established a clear chain of evidence between raw data, emerging concepts, and aggregate dimensions. A data table supporting emergent concepts is shown in Appendix B.

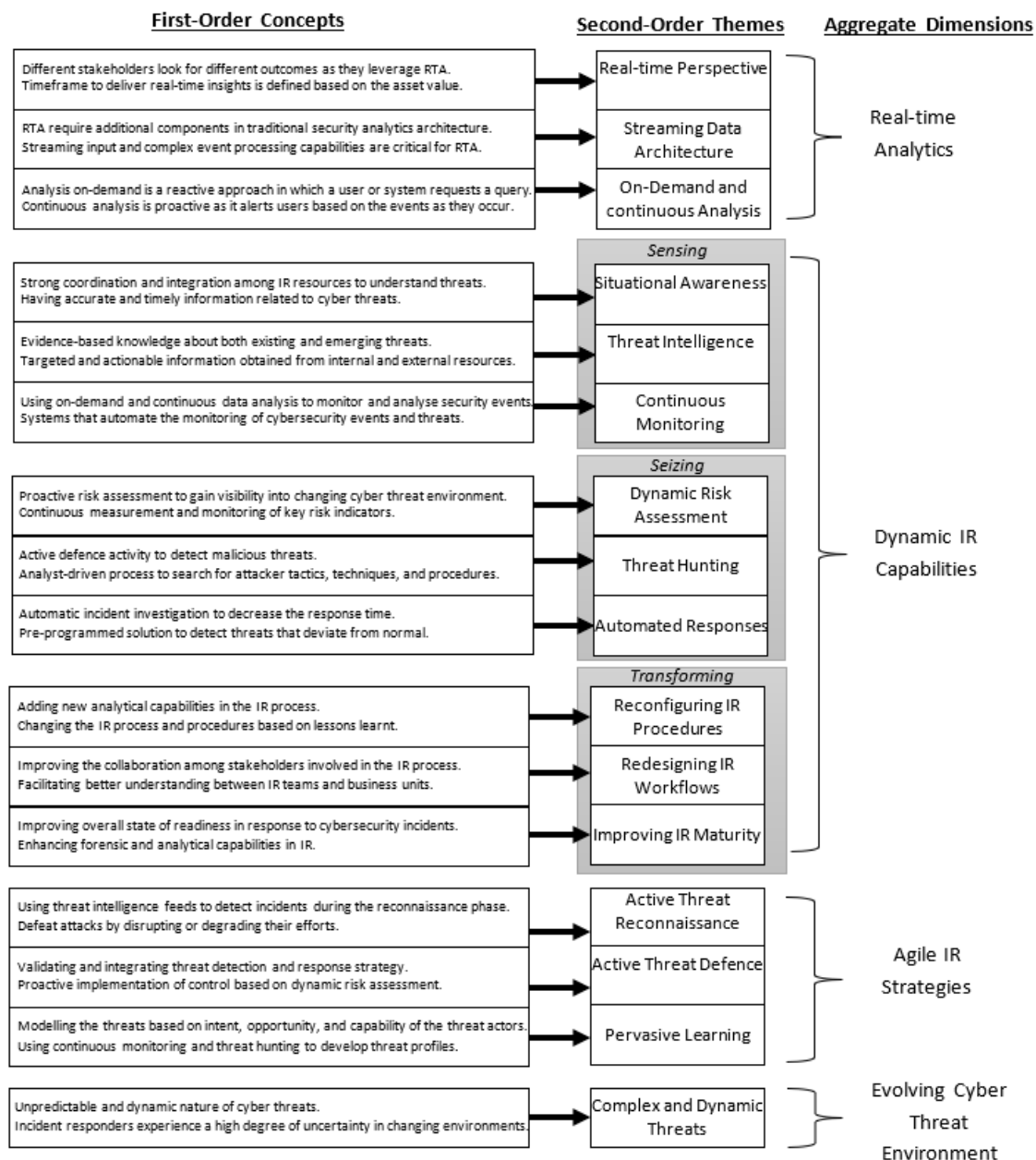


Figure 1. Data structure

4. Emergent findings

We now present and provide evidence for the findings that emerged during the analysis of the case studies.

4.1. Evolving cyber threat environment

Each of the case organisations experienced an evolving cyber threat environment characterised by a diverse range of cyber threats, attack vectors, and threat actors. FinInsuranceA's General Manager of Cybersecurity Strategy and Governance noted:

We are targeted not only by skilled hackers and financial criminals but also well-organised threat actors, including professional criminals, hacktivists groups driven by social or political agendas, insiders, and state-level actors ... we also face diverse cyber threats such as advanced persistent threats [APT], phishing, distributed denial of service, spear-phishing, SQL injection and zero-day attacks.

As the focal organisations continuously invested in business and technology innovations, new complexities and vulnerabilities were introduced into their financial services technology ecosystem. FinBank's Cybersecurity Senior Manager explained that:

The adoption of cloud, mobile, social media technologies, and focus on outsourcing is blurring the boundaries of the technology ecosystem. This has resulted in [a] much broader "attack surface" for the threat actors to exploit.

The diverse range of threat actors, motives, and attack methods compelled focal organisations to develop RTA capability in the IR process. FinInsuranceA's General Manager of Cybersecurity Strategy and Governance noted:

Real-time analytics helps our cyber defence and response team to analyse events as they occur and then determine if the event is really an incident. An event is just an item of interest, and an incident is something that has caused a loss or an impact, or it is about to or has the potential to cause a loss or impact. If they are an incident, then we respond to them using our incident and response processes.

4.2. Building dynamic capabilities to enable agility in incident response: a theoretical framework

In this section, we present a framework (see Figure 2) that explains how the case organisations built dynamic IR capabilities using RTA and thereby enabled agility in their IR process. Our framework is grounded in RTA practice in the IR process at the case organisations. The complex and evolving nature of the cyber threat environment triggered the organisations to use RTA in their IR process. Specifically, they used the *real-time perspective*, *streaming data architecture*, and *continuous and on-demand analysis* features of RTA capability to develop microfoundations that underpinned the building of sensing, seizing, and transforming dynamic IR capabilities. These dynamic IR capabilities in turn helped them to enable agility in IR by leveraging innovative, flexible, and swift IR strategies such as *active threat reconnaissance*, *active threat defence*, and *pervasive learning*.

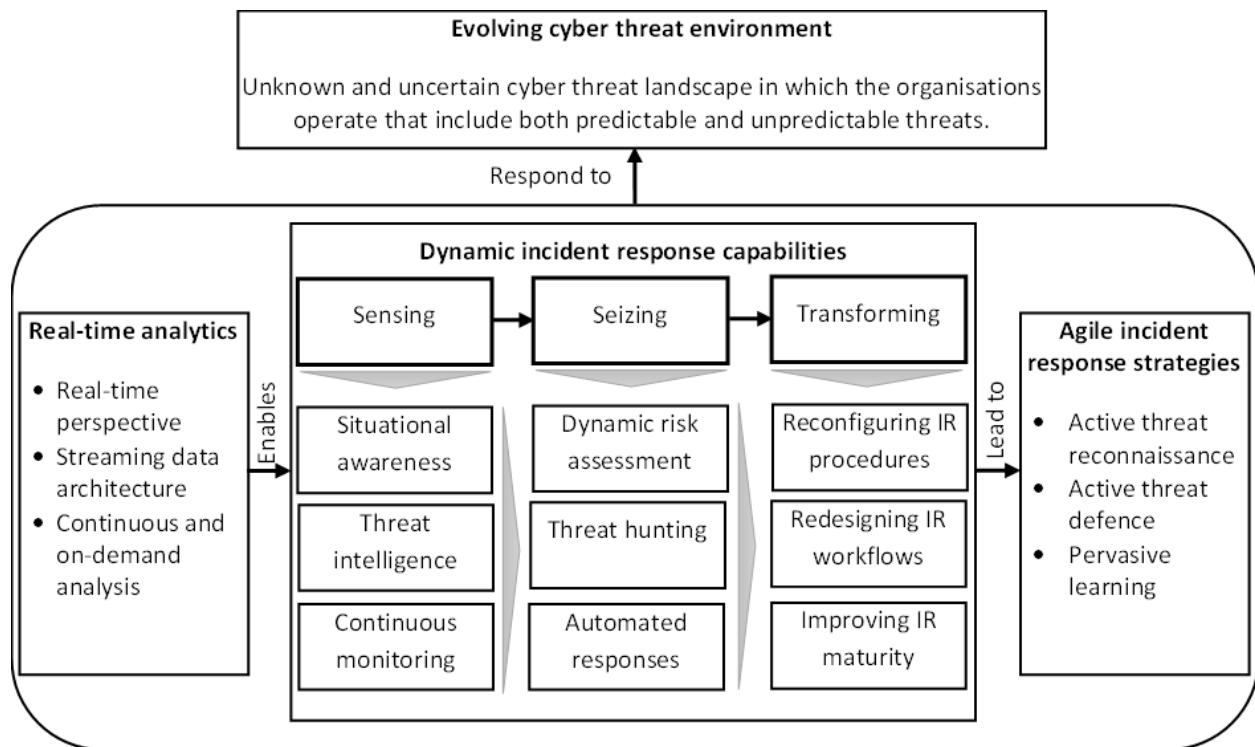


Figure 2. Building dynamic capabilities to enable agility in IR using RTA: A theoretical framework

4.3. Real-time analytics capability in cybersecurity incident response

It was evident early on in our research that participants understood the term “real-time analytics”

in diverse ways. Therefore, defining *real-time perspective* in terms of what RTA means and what the timeframe is to deliver real-time insights into the IR process was especially important.

FinInsuranceA's General Manager of Cybersecurity Strategy and Governance noted:

Real-time analytics has [a] different meaning to different people. It depends on your role in the incident response process ... The important thing to consider is what data sources are relevant for real-time data analysis and what outcome we want to achieve using real-time analytics.

The Head of Global Insider Threat Detection in FinInsuranceB discussed the timeframe to deliver real-time insights as follows:

Real-time analytics is about taking the streaming data in and processing it to generate insights within the timeframe that we have agreed on and meets our requirements.

The critical factor to consider in defining the timeframe for RTA is to evaluate the asset value. When the most important organisational assets are under attack, then it is critical to execute the lowest latency response. FinBank's General Manager of Cybersecurity Risk Management stated:

We define the real-time analytics timeframe depending on the asset value and how important it is to execute an appropriate response once a threat is detected. So, when we are trying to stop a cybersecurity attack or compliance threat before it causes any damage to us, it is critical that we execute the lowest latency response.

Traditionally, the architecture for most cybersecurity analytics solutions is designed on the logic of batch processing of data, in which data from different sources such as intrusion detection systems, firewall logs, network flow logs, antivirus, and spam logs are extracted, transformed, and then loaded into a centralised repository such as a SIEM system.

FinInsuranceB's Chief Security Architect noted that RTA capability required well-designed *streaming data architecture* to ingest, process, and analyse streaming data from multiple sources and also some additional components as follows:

Cybersecurity analytics requires an architecture that can handle data from multiple sources in greater volume than at present, integrate this data into a centralised repository [SIEM] and then provide insights that can lead us quickly to the most pressing issues. When we started building real-time analytics capability, we needed to add real-time analytics tools and design streaming data architecture to analyse security logs in real-time. So, for example, some of the logs that we need in the SIEM also go into the real-time analytics platform as streaming data, where it can be used for real-time analytics ... For that, we added several additional components in our analytics architecture to implement real-time analytics such as in-memory analytics, complex event processing, data virtualisations and service-oriented architecture.

SIEM tools were identified as an integral part of streaming data architecture and a key enabler of RTA in focal organisations. SIEM tools enable real-time search, monitoring, and analysis of cybersecurity events that can be fed through dashboards to update IR teams about what is happening across the cyber threat environment. FinInsuranceA's General Manager of Cybersecurity Strategy and Governance stated:

The tool that we are using to enable real-time analysis in our cybersecurity incident response process is SPLUNK. What it enables us to do is real-time analysis, search, and monitoring of cybersecurity events ... So, we are able to drill down and monitor in real-time using different views across our dashboards.

To generate useful insights related to cybersecurity incidents and threats, the focal organisations used two specific types of RTA: *on-demand* and *continuous analysis*. FinBank's General Manager of Data and Analytics differentiated these two types as follows:

Analysis on demand is a reactive approach in which a user or system requests a query to further investigate and understand what is happening right now ... whereas continuous data analysis is a proactive approach that alerts users or triggers actions based on the events as they occur.

The participants also gave examples of how they use both on-demand and continuous data analysis to combat advanced threats by analysing network and log data from security systems and integrating it with external threat intelligence feeds to generate comprehensive cybersecurity insights. FinInsuranceB's Director of Global Cyber Forensics noted:

To combat advanced cybersecurity threats, such as sensitive data exfiltration and zero-day malware, we use both on-demand and continuous real-time analytics to analyse all network traffic, log data from applications, security systems, and network data and also fuse external threat intelligence in real-time.

In the next section, we explain how key features of RTA helped focal organisations to develop dynamic IR capabilities.

4.4. Dynamic incident response capabilities

The key features of RTA in the IR process enabled nine RTA-based microfoundations that underpin the building of sensing, seizing, and transforming dynamic IR capabilities. The details of these dynamic IR capabilities are given below.

4.4.1. Building sensing capabilities in incident response

We noted three microfoundations of sensing capabilities in IR: situational awareness, threat intelligence, and continuous monitoring.

All participants highlighted that real-time *situational awareness* is critical to understanding what is happening in the cyber threat environment and how an incident will unfold. That is why the key stakeholders in the IR process, including the chief information security officer, cybersecurity managers, and cybersecurity analysts in focal organisations, used RTA to develop real-time situational awareness about the malicious activities in their cyber threat environment. FinBank's General Manager of Data and Analytics noted:

Situational awareness relies on real-time data and analytics. For example, using stream processing of data from network flows, sensors, and metadata, we look for suspicious or anomalous network activities that strongly indicate a security incident is in progress.

FinInsuranceA's Manager of the Cyber Defence Centre further explained how they used dashboards to monitor key information regarding cybersecurity events to maintain awareness of the cybersecurity environment:

Detecting and understanding potential cybersecurity threats to our business is not enough. Information from monitoring tools and threat intelligence needs to be consolidated and communicated to [cybersecurity] managers ... The situational awareness provided by the consolidated information in the dashboards enables us to understand what's happening and respond to incidents without wasting any time.

To build sensing capabilities in strengthening situational awareness, participants emphasised the importance of *continuous monitoring* and real-time *cyber threat intelligence* on both external and internal cyber threats. FinInsuranceA's Head of Cyber Threat Detection and Response Team explained how they used continuous monitoring systems to gain situational awareness:

Our continuous monitoring systems run all day and monitor events as they occur and generate alerts as soon as a threat is detected that needs a response. The response can either come from our incident response team or from the system itself.

FinInsuranceA's General Manager of Cybersecurity Strategy and Governance explained that the ability to analyse data and extract anomalies that may indicate an active threat in the network is another example of scanning for cyber threats using continuous monitoring and real-time threat intelligence:

On an operational point of view, we also have processes where we are proactive in addressing the [cyber] threat landscape. We have continuous threat intelligence feeds from networks in which we use analytics to analyse what is happening and detect anomalies.

FinInsuranceA's Manager of Cybersecurity Strategy and Governance also discussed how their managed services analytics team used real-time threat intelligence to detect email malware:

Our managed services provider has analytics visibility across multiple clients. As we are one client, they can analyse to see if it is a zero-day attack. It is an email malware, but we have actually seen this trend across multiple clients. So, we get an early warning. We have got maybe 20 such emails, and we can pull those from the inbox. Our response is to the point where we can contain it so that there is no impact. This is because we now know that the

email is a bad email, and we can now actually withdraw that from user mailboxes before they even open the email and click the attachment.

4.4.2. *Building seizing capabilities in incident response*

We noted three microfoundations of seizing capabilities in IR: dynamic risk assessment, threat hunting, and automated responses.

Focal organisations were increasingly adding proactive approaches to respond to cybersecurity incidents dynamically, along with reactive approaches to increase the visibility of their cybersecurity environment. The participants highlighted *dynamic risk assessment* as the critical dynamic capability to seize on the latest threat intelligence and thereby become proactive in cybersecurity IR. FinInsuranceA's General Manager of Cybersecurity Strategy and Governance noted:

In my own team's roadmap, the priority is to improve the measurement of metrics and report the coverage and effectiveness of our security controls and do dynamic risk assessment by looking at what we call Key Risk Indicators.

FinInsuranceA's Manager of Cybersecurity Strategy and Governance also noted two types of key risk indicators they were looking at—lag indicators and lead indicators:

What we are looking at is what we call lag indicators and lead indicators. So, historically, we are very good with lag indicators ... Lag indicators provide statistics around (a) this is how much spam we received last month, (b) this is the number of desktops that have been affected by malware, and (c) this is how many malicious URLs that were accessed. But they are after the event ... They tell us something bad that happened which we detected. What lead indicators tell us is how things are working and can give us some information about the future so that we can take pre-emptive actions. So, for example, the lead indicator for us can be the number of people that requested a security exemption to formally be noncompliant with a security policy or standard. This indicates to us that maybe our risk appetite is invalid. We need to change our risk appetite, or that we need to do more work around awareness or education.

FinBank's General Manager of Data and Analytics compared batch processing of data and analytics with real-time analytics. He explained that analytical applications that process data

in batches deliver historical insights based on lagging indicators. In contrast, the insights generated by real-time analytical platforms are more current and predictive in nature because they are based on leading indicators:

If I make a comparison of using batch style analytics with real-time analytics, it may take hours or even days to yield results using batch style analytics. So, our batch analytical applications collect and correlate data from multiple sources such as intrusion detection systems, SIEM, spam and antivirus software, and network and application logs so that we can gain visibility into most pertinent insights related to cybersecurity risks and can make informed actions accordingly and therefore [we] are very good in delivering “after the fact” insights (lagging indicators). In contrast, the insights that we get from analysing leading key risk indicators are predictive in nature and allow us to get ahead of the curve.

To build up a dynamic risk assessment capability, the participants emphasised the importance of *threat hunting* and *automated responses*, providing examples of both in the context of enabling proactive IR. Threat hunting was used to identify previously invisible threats and to gain an understanding of the organisation’s risk exposure. FinBank’s Head of the Cybersecurity Incident Response Team noted:

Threat hunting is the process of proactive and iterative detection, isolation, and mitigation of advanced threats that are not detected by traditional security controls ... with real-time threat hunting, we get forward-looking visibility and continuous monitoring for patterns of attacker activity.

Automated responses helped in generating automatic threat alerts and triggering actions based on predefined business rules. A FinBank Cybersecurity Senior Manager noted:

We are facing new types of cybersecurity threats every day, and the most significant way to combat ... them is through intelligent and automated defence mechanisms that can quickly identify existing and new threats and then ... generate alerts and trigger [automated] actions to mitigate them ... developing such automated response capability required reshaping and even retooling of our cybersecurity infrastructure.

However, according to FinInsuranceB's Head of Global Insider Threat Detection, the cybersecurity IR process cannot be fully automated and requires the involvement of cybersecurity managers to investigate, analyse, and confirm threats, particularly when they are dynamic and unpredictable (such as malicious insider threats):

Given the sensitive nature of cybersecurity issues, the cybersecurity incident response process cannot be fully automated, and security managers still need to investigate and confirm threats, particularly when the threats are internal.

Therefore, an automated response is one of the key outputs of RTA in the cybersecurity IR process. The goal of using automation in IR is to swiftly detect and respond to cybersecurity incidents. However, human involvement (cybersecurity managers) is critical, because the complexity of IR requires human input during the process of assessing the impact, scope, and severity of cyber-attacks.

4.4.3. Building transforming capabilities in incident response

We noted three microfoundations of transforming capabilities in IR: reconfiguring IR procedures, redesigning IR workflows, and improving IR maturity. A prominent theme that emerged while analysing transforming capabilities in IR was the discussion around strategic issues related to IR procedures, workflows, and maturity. For example, FinInsuranceB's Chief Security Architect stressed that *improving IR maturity* to support RTA was essential to enhance their state of readiness in responding to cybersecurity incidents swiftly and effectively:

Identifying what our current level of capability maturity is helps us to identify what our gaps are. We can then formulate our strategy to fill these gaps and improve our capability for cybersecurity services over time. We translate that into an actionable roadmap of investment in technology, processes, people, and skills development to try and achieve that outcome to continuously improve our detection, protection, response, and recovery capabilities to protect our information systems and assets.

To enable IR maturity, participants emphasised the importance of *reconfiguring IR procedures* and *redesigning IR workflows* to add new analytical capabilities. For example,

FinInsuranceA's Manager of Cybersecurity Strategy and Governance discussed how they reconfigured their IR workflows and procedures to develop shared threat intelligence capability in their IR:

We also participate in threat intelligence sharing communities. So, through our participation in these forums and collaboration on threat intelligence feeds that we have, we can also analyse and then take proactive countermeasures before those attacks are actually experienced by our organisation. So, on a daily basis, we are getting feeds on things like trustworthy IP addresses and domain names. We are getting feedback from our peers across [the] industry around phishing campaigns and new emerging malware. Then our cyber defence and response team is proactively taking and analysing that information and then translating that into changes in our security services to try and provide a countermeasure before we actually see those attacks hit our enterprise.

Participants also explained how they improved the collaboration between the analytics and IR teams by redesigning IR workflows. For example, FinBank's Head of Cybersecurity Incident Response Team stated:

The first step we did to improve the analytical capability in [the] incident response process was to collaborate and establish a relationship with our analytics team. We started embedding some of their practices and processes and even using some of their services. By building stronger relationships and leveraging the capabilities of the analytics team, we were able to improve the analytical workflows and drive our decisions based on the data.

In the next section, we explain how the aforementioned dynamic IR capabilities empowered the focal organisations to enable agility in the IR process.

4.5. Agile incident response strategies

The most challenging aspect of the IR process in focal organisations was the timely and accurate detection and assessment of possible cybersecurity incidents, followed by the selection and execution of the most appropriate response strategy. Traditional IR processes and strategies were not sufficient to deal with the evolving cyber threat landscape. Thus, focal organisations used dynamic IR capabilities to implement agile IR strategies, such as *active threat reconnaissance*,

active threat defence, and *pervasive learning*.

In *active threat reconnaissance*, focal organisations leveraged real-time situation awareness and threat intelligence feeds to detect some cybersecurity incidents during the reconnaissance phase, prior to being attacked. FinBank's Chief Information Security Officer noted:

Our incident response team uses cyber threat intelligence feeds to understand the procedures, tactics and techniques of the attackers and can stop some attacks by degrading or disrupting their efforts. In this way, cyber threat intelligence helps us in detecting an incident during the reconnaissance phase; that is, before we have actually been attacked.

In *active threat defence*, focal organisations used dynamic risk assessment to proactively implement a spectrum of cybersecurity measures that secured their critical data and systems from malicious threats. FinInsuranceA's Manager of Cybersecurity Strategy and Governance explained:

Sometimes people commonly misunderstand proactive may necessarily be equal to protective or preventative, and yet if you talk to a lot of experts in the cybersecurity industry, current thinking basically says that you can assume that at some point you might actually be compromised, or you have a breach of some sort. So rather than just saying by putting all your investment in a preventative approach, you can still be proactive in boosting your detection and response capabilities so that when something happens you know that sooner, which means you can decrease the impact and then also recovery and all that is well planned so that you cater better in reduction of impact.

In *pervasive learning*, focal organisations leveraged continuous monitoring and threat hunting to learn about the intent, capability, and opportunity of the threat actors and develop a detailed threat profile. FinInsuranceB's Director of Global Cyber Forensics noted:

Modelling the threats based on intent, opportunity, and capability of the threat actors help[s] us to develop a comprehensive organisational threat profile that includes information about the threat scenarios, threat actors, and critical assets.

FinBank's Security Operations Centre Manager explained how they had designed their cybersecurity IR process around the cyber kill chain model (an industry standard) that helped them to focus on the different stages of an attack and thereby execute an agile response strategy:

We have designed our cybersecurity incident monitoring and response plan based on [the] cyber kill chain model that identifies the sequence of stages which an attacker must pass through to reach the desired goal. The incident response strategy and execution of recommended actions from the playbook are dependent on the stage where [the] attack was detected.

Appendix C provides an illustrative example of agile IR in action by explaining how FinBank's IR team detected and discerned diverse types of phishing attacks and responded to them in an agile manner using RTA-based sensing, seizing, and transforming dynamic IR capabilities.

5. Discussion

5.1. Contributions to the literature on cybersecurity IR: demystifying the black box of cybersecurity IR agility

Here, we address calls to demystify the black box of cybersecurity IR agility (Baskerville et al., 2014; A. Naseer et al., 2021; Siregar & Chang, 2019), in particular, through the implementation of agile IR strategies in response to evolving cyber threats by developing dynamic capabilities.

Unlike previous research that utilises dynamic capabilities and its microfoundations (sensing, seizing, and reconfiguring) to develop a data breach recovery framework based on the secondary data (Shankar & Mohammed, 2020), our study presents a framework based on empirical findings that demystifies the black box of enabling agility in IR by linking RTA with IR agility through the development of dynamic IR capabilities—a link that is poorly understood to date. We extend existing cybersecurity IR research in the following ways.

Our findings provide new insights into how RTA-based microfoundations underpin the development of sensing, seizing, and transforming dynamic IR capabilities, which, in turn, help organisations to enable agility in IR. Interestingly, the literature has yet to fully report on the fact

that organisations need to build sensing capabilities in IR to deal with the evolving cyber threat environment (Goel et al., 2023). We report that sensing in IR consists of microfoundations relating to situational awareness, threat intelligence, and continuous monitoring. Organisations that continuously monitor their cyber threat environment and generate shared threat intelligence using sensing capabilities can build strong situation awareness and reduce their vulnerability. Through the framework, we explain how IR teams use novel cybersecurity and analytical solutions, such as SIEM, complex event processing, and continuous monitoring tools, to make sense of the evolving cyber threat environment in real time that previously was not possible. Our data suggest that sensing dynamic capabilities in IR, such as shared threat intelligence and situational awareness, are essential for the swift detection of malicious threats in the evolving cyber threat environment. In this way, our findings resonate with research that reports that shared threat intelligence and situational awareness provide several benefits to organisations, such as improved learning and better decision-making in IR (Shin & Lowry, 2020; Webb et al., 2014).

Additionally, our findings help to explain how IR agility is manifested by supporting organisational-level strategic tasks of sensing and responding to internal and external cybersecurity threats as they occur. By doing so, we extend the existing insights (He et al., 2022; Lee et al., 2015; Overby et al., 2006; Park et al., 2017) by showing how IR agility is a manifested type of dynamic capability that enables an organisation to adjust its existing IR processes and procedures or create new ways of detecting, preventing, and responding cyber threats to deal with the changing cyber threat environment. Our findings show that the heterogeneous nature of threat actors, motives, and attack methods is pushing organisations to use RTA to enable agility in IR and align with existing research showing that IR agility is central for operating in the evolving cyber threat environment (Kotsias et al., 2022; A. Naseer et al., 2021). Our findings emphasise that organisations require seizing capabilities that ensure the mobilisation of IR resources and capabilities to thoroughly investigate and respond to diverse threats. We found that seizing in IR consists of microfoundations relating to dynamic risk assessment, threat hunting, and automated

responses. Given the complex and dynamic nature of cyber threats, organisations continuously innovate with IR methods to build seizing capabilities that swiftly mobilise their IR resources and capabilities for agile responses to malicious threats.

Unlike previous research that concentrates on the adoption of an agile manifesto to address agility challenges in IR (Grispos et al., 2017, 2014; He et al., 2022), our data indicate that organisations need transforming capabilities to enable agility in IR. Transforming capabilities in IR consist of microfoundations relating to reconfiguring IR procedures, redesigning IR workflows, and improving IR maturity. Through our framework, we show that transforming capabilities in IR support organisations with: (1) continuous renewal of IR assets; (2) redesigning flexible and innovative IR workflows to improve collaboration during the IR process both externally and internally; (3) reconfiguration of the IR procedures to add new analytical capabilities; and (4) improvement of the overall maturity of the IR process. Moreover, transforming IR capabilities also help organisations in building analytics-driven IR strategies that provide quick and agile responses to changes in the cyber threat landscape. In this way, our findings align with existing research on dynamic capabilities showing that transforming capabilities support organisations with the continuous strategic renewal of assets and organisational structures to ensure responsiveness in fast-changing environments (Conboy et al., 2020; Teece, 2014; Warner & Wäger, 2019).

5.2. Contributions to the literature on dynamic capabilities: from value creation to value protection strategies

We contribute to the dynamic capabilities literature by contextualising this theoretical lens to the specific case of cybersecurity IR. Dynamic capabilities has offered us a powerful conceptual and theoretical framework to explain how organisations alter their resource base—acquire and shed resources, integrate them, and recombine them—to generate new value-creating strategies (Eisenhardt & Martin, 2000). Thus far, this literature has mainly articulated how these value-creating strategies are essential if the firm is to sustain competitive advantage in the longer

term as customers, competitors, and technologies change (Schilke & Helfat, 2018; Steininger et al., 2022; Teece et al., 2016). As evident by our findings, in the context of cybersecurity, value protection strategies are essential if the firm is to safeguard competitive advantage by protecting its critical enterprise systems and data assets in the rapidly evolving cyber threat environment. Our framework identifies nine RTA-based microfoundations that underpin the development of sensing, seizing, and transforming dynamic IR capabilities. These, in turn, enable organisations to implement value protection strategies such as active threat reconnaissance, active threat defence, and pervasive learning.

The application of the microfoundations of sensing, seizing, and transforming dynamic capabilities perspective in the cybersecurity context also requires consideration of several observations such as those catalogued by Teece (2007) and Teece et al. (2016). First, Teece (2007, p. 1322) notes that sensing (and shaping) new opportunities (and threats) is very much a scanning, creation, learning, and interpretative activity that analyses diverse information about trends in the business ecosystem. In the cybersecurity context, where organisations use the IR process to respond to complex and evolving cyber threats, organisations need *sensing* capabilities to scan their environment so that they can detect threats before they cause any damage (H. Naseer et al., 2018; Shankar & Mohammed, 2020). Our findings show that sensing in IR is very much a scanning, learning, and interpretive activity that analyses diverse information about cyber threats in the business environment. Second, Teece et al. (2016, p. 18) argue that to address opportunities or neutralise threats, incumbents require seizing capabilities that allow the mobilisation of resources to address needs and opportunities, and *create value* from doing so. As our findings show, to *protect value*, organisations need seizing capabilities that ensure the mobilisation of IR resources and capabilities to thoroughly investigate any malicious activity and to neutralise cyber threats. Third, sensing and seizing capabilities help create and discover opportunities; however, it is the transforming capabilities that allow the continuous strategic renewal of assets and organisational structures to ensure responsiveness in fast-changing

environments (Teece et al., 2016). In the context of cybersecurity, our findings suggest that sensing and seizing capabilities help to identify, investigate, and neutralise cyber threats. However, to ensure responsiveness against cyber-attacks in the rapidly evolving cyber threat environment, organisations need transforming capabilities that enable continuous alignment and realignment of IR resources and capabilities.

5.3. Contributions to the literature on BA: using RTA to enable agility in IR

We contribute to the BA literature by identifying the specific dimensions of RTA capability that enable agility in the IR process. These dimensions include 1) *defining the real-time perspective*, 2) *building streaming data architecture*, and 3) *on-demand and continuous data analysis*. While previous research in IR agility identified how RTA enables agility in the cybersecurity IR process and thereby improves enterprise cybersecurity performance (A. Naseer et al., 2021), we extend this by providing details about how organisations use RTA to develop dynamic IR capabilities that indirectly contribute to enterprise cybersecurity performance.

The term “real-time” has many perspectives or meanings in terms of the timeframe to deliver real-time insights. That is why the terms “near real-time”, “right time” and “true real-time” are used interchangeably to reflect the timeframe of RTA (Russom et al., 2014; Townsend et al., 2018). Our findings suggest that the real-time perspective is the timeframe within which IR teams need to deliver a response to cybersecurity incidents before they can cause any damage. In addition, our findings emphasise that the timeframe to generate real-time insights depends on the asset value and the stakeholder’s role in the IR process. RTA as an innovative technology requires architectural changes in the traditional security analytics architecture (A. Naseer et al., 2021; Phillips-Wren et al., 2015). Our study provides empirical evidence for this observation and suggests that RTA practices in IR require a well-designed streaming-data architecture that enables analytical capabilities such as streaming input and data analysis, complex event processing, and continuous monitoring.

5.4. Implications for practice – three pillars in understanding how to enable agility in IR

The findings of our study have considerable practical significance and contribute to IR practice in the following ways. First and foremost, our theoretical framework and analysis point to three main pillars that help IR practitioners understand the importance of building dynamic IR capabilities using RTA and how they lead to the implementation of agile IR strategies. To this end, it is critical to understand the nuanced role of RTA in the IR process. Rather than using RTA at an operational level only to continuously monitor the cyber threat landscape (A. Naseer et al., 2021), cybersecurity managers can use RTA in a more innovative way to build strategic-level dynamic IR capabilities. Explicit consideration of the system of dynamic capabilities enabled by RTA in the IR process will help cybersecurity management to appropriately evaluate technology investments in this area. Furthermore, our findings indicate that the development of real-time decision automation systems using RTA enables agility in IR. Organisations that leverage RTA-enabled decision automation systems detect and respond to cybersecurity incidents more swiftly than those running entirely on manual processes. The utilisation of RTA-enabled decision automation in IR will help cybersecurity managers yield better performances from their IR teams because they can devote their time to more important and less repetitive cybersecurity tasks. However, our findings show that the deployment of such systems requires careful consideration and customisation to ensure that they are tailored to the specific needs and requirements of the organisation.

Second, given the dynamic and evolving nature of the cyber threat environment, we argue that it is very important for organisations to continuously learn from cybersecurity incidents. IR teams can use our framework to minimise reactivity and enable continuous learning in the IR process by leveraging agile IR strategies. RTA-based sensing, seizing, and transforming dynamic IR capabilities, taken together as a system of capabilities, facilitate the implementation of agile IR strategies such as active threat defence, active threat reconnaissance, and pervasive learning in the IR process. Continuous learning from past cybersecurity incidents and swift detection of

emerging cyber threats will help IR teams to build cyber resilience by swiftly detecting and isolating future cybersecurity incidents, containing the damage, and recovering from the attack, while maintaining continuous business operations (Goel et al., 2023).

Third, we propose that vendors who develop cybersecurity and RTA solutions should recognise the far-reaching transformational role that their solutions may provide to organisations. Building cybersecurity solutions that can (1) automate investigations and forensic analyses, (2) quickly integrate threat intelligence feeds, and (3) apply visual analytics and complex algorithms to discover the potential threats will help clients to enable agility in their IR processes by executing proactive IR strategies that can better deal with the evolving cyber threat environment. However, our findings show that the significant potential of dynamic IR capabilities enabled by RTA will also increase the demand for data integration, analytics, automation, and visualisation. Thus, vendors who develop cybersecurity and RTA solutions need to carefully consider these requirements when developing their solutions.

5.5. Limitations and future research

The context of this study raises questions about the generalisability of the proposed framework and suggests possibilities for future research. The generalisation of our findings in other contexts that involve the adoption of RTA should be applied with caution because the findings may be particular to the specific characteristics of the IR processes and sites studied. For example, we acknowledge that the selection of financial sector organisations, the use of similar IR strategies in the organisations studied, as well as their high levels of maturity in adopting RTA, may limit the generalisability of our findings. Future research is needed that would extend, support, or reject our findings in other organisations and industry sectors. In addition, given the small number of cases in our study, the probability is high that relevant dynamic IR capabilities other than the ones observed in our cases exist that can enable agility in IR using RTA. Therefore, researchers are encouraged to further explore the role of dynamic capabilities in enabling agility in IR in

other contexts and organisational settings. Despite these limitations, this research suggests that RTA practices in the cybersecurity IR process open a new direction for IR research that considers the implications of BA-enabled dynamic capabilities in implementing agile IR. The findings from this study will serve as a basis for future research that can be undertaken to challenge, confirm, and extend this study's conclusions.

Future research is needed to further investigate the conditions that facilitate or hinder the development of dynamic IR capabilities. In our study, the IR practices within each organisation were structured within a dedicated SOC that used SIEM tools and security analytics solutions to identify and manage cyber threats. Future research could investigate different structures of IR practices in organisations and whether the structure or the analytics tools have an impact on the development of dynamic IR capability.

Finally, linking agility through dynamic capabilities and organisational performance is the subject of ongoing research. Further research is also needed to identify key differences between RTA and other disruptive technologies to gain insights into how RTA may offer distinct capabilities. Because the adoption of RTA facilitates a paradigm shift in the IR decision-making process, more detailed studies are required to investigate its potential as well as the challenges it poses to organisations, both large and small.

6. Conclusion

In the increasingly complex and evolving cybersecurity environment, organisations face new types of threats every day and they need to detect and respond to them quickly—ideally before the threats materialise. We studied RTA practice in the IR process at three large financial organisations. Grounded in our data, we derived a framework that explains how organisations respond to the evolving cyber threat environment by developing dynamic IR capabilities using RTA. These dynamic IR capabilities allow organisations to enable agility in their IR process and confront cybersecurity threats with agile IR strategies, such as *active threat defence*, *active threat reconnaissance*, and *pervasive learning*. Our framework explains in detail how RTA-based

dynamic IR capabilities minimise reactivity and thus enable IR teams to take a more proactive approach while responding to both predictable and unpredictable cyber threats. The framework has several implications for both theory and practice.

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Appendix A: Examples of open-ended questions that guided the data collection

Research Question: How do organisations enable cybersecurity incident response agility by developing dynamic capabilities using real-time analytics?

Methodology: Multiple case studies

Main Data Collection Method: Semi-structured interviews

Other Data Sources: Organisational documents, strategic progress reports, internal publications, and field notes

Interview Questions (Excerpt)

Theme 1: Real-Time Analytics in Cybersecurity Incident Response

- What does RTA mean to you?
- Describe the evolution of analytics in your organisation?
- Describe how you have integrated RTA into your incident response process.
- What are the sources of data for RTA? How do you manage your data and analytical architecture?
- What kind of reporting and analysis are you performing on cybersecurity data related to risk management and incident response? What type of analytical models are you using?

Theme 2: Impact of Using Real-time Analytics on Cybersecurity Incident Response

- Describe the incident response process from initial identification to closure.
- Describe how important it is to be proactive and dynamic when it comes to executing incident response?
- How do you generate cyber threat intelligence using RTA?
- What are some of the key risk indicators that you monitor, analyse, and measure using RTA and why?

Theme 3: Agility in Cybersecurity Incident Response

- What specific benefits have you realised by using RTA in your cybersecurity incident response process?
- What role does RTA play in becoming proactive/dynamic in incident response?
- Give a few examples of innovative ideas that you have incorporated into your incident process that have resulted in better execution of your incident response process?

Appendix B: Example of data supporting interpretations of second-order themes

Aggregate dimension	Second-order themes		Representative quotes
<i>Real-time analytics</i>		Real-time perspective	"We use different types of approaches to address different types of cybersecurity incidents. As there is no single plan that fits for all purpose, our cybersecurity incident response timeframe can range from few minutes up to few hours." (Manager of Cybersecurity Strategy and Governance, FinInsuranceA) "We use real-time analytics to determine whether an incident requires attention right now and which assets are impacted by the incident." (Head of Cybersecurity Incident Response Team, FinBank)
		Streaming data architecture	"The tool that we are using to enable real time analysis in our cybersecurity incident response process is SPLUNK. What it enables us to do is real-time analysis, search and monitoring of cybersecurity events...So, we are able to drill down\monitor in real time using different views across our dashboards." (General Manager of Cybersecurity Strategy and Governance, FinInsuranceA) "We have added several additional components in our analytics architecture to implement real-time analytics such as in-memory analytics, data virtualisations and service-oriented architecture." (Chief Security Architect, FinInsuranceB)
		Continuous and On-demand analysis	"Real-time analytics helps us to react to cybersecurity threats without any delay...In simple words, what we are doing is continuously analysing cybersecurity events to detect potential threats and prevent incidents before they happen." (Manager of Cybersecurity Strategy and Governance, FinInsuranceA) "To combat advanced cybersecurity threats, such as sensitive data exfiltration and zero-day malware, we use both on-demand and continuous real-time analytics to analyse all network traffic, log data from applications, security systems, and network data and also fuse external threat intelligence in real-time." (Director of Global Cyber Forensics, FinInsuranceB)
<i>Dynamic IR capabilities</i>	S e n s i n g	Situational awareness	"Because we are using continuous monitoring systems to analyse cybersecurity events across the network, this greatly improves the level of cybersecurity awareness of our cybersecurity managers." (Security Operations Centre Manager, FinBank) "To achieve successful continuous monitoring of networks and improve our cybersecurity awareness, both detective and proactive monitoring actions must work together." (Head of Global Insider Threat Detection, FinInsuranceB)
		Threat intelligence	"Our cyber defence and response team is proactively taking and analysing that information and then translating that into changes in our security services to try and provide a counter measure before we actually see those attacks hit our enterprise." (Cybersecurity Senior Manager, FinBank) "We have threat intelligence feeds in which we use analytics to analyse what is happening and we also participate in threat intelligence sharing communities." (Senior IT Risk Partner, FinInsuranceA)
		Continuous monitoring	"Getting an understanding of what to monitor, where to monitor and how to monitor is very important. Continuous monitoring does not mean that everything including all applications, systems, end points, networks, security processes and infrastructure needs to be monitored everywhere and all the time. Therefore, it is very important to determine what needs to be monitored and set monitoring policies around those needs." (Chief Information Security Officer, FinBank)
	S e i z i n g	Dynamic risk assessment	"By consolidating information from continuous monitoring systems into a dashboard, our threat hunting team can identify and monitor assets that have higher levels of risk and then respond to threats against these assets appropriately." (Head of Cybersecurity Incident Response Team, FinBank) "We are definitely more interested in reporting and analysing the lead indicators in real-time that would demonstrate that the organisation is drifting outside of its risk appetite." (Director of Global Cyber Forensics, FinInsuranceB)
		Threat hunting	"Threat hunting is the process of proactive and iterative detection, isolation, and mitigation of advanced threats that are not detected by traditional security controls." (Head of the Cybersecurity Incident Response Team, FinBank) "If I make a comparison of using batch style analytics with real-time analytics, it may take hours or even days to yield results using batch style analytics. Our batch analytical applications are very good in delivering 'after the fact' insights [lag indicators]. In contrast, the insights that we get from analysing lead key risk indicators allow us to get ahead of the curve." (Manager of IT and Information Security, FinInsuranceA)
		Automated Responses	"The progression to automated incident response in real-time is often gradual. What I mean by that is, the improvement in the speed with which we executed the response was improved gradually, such as getting from days to hours to minutes or even seconds. Then, if required further refinement can also be done based on the business need." (General Manager of Cybersecurity Risk Management, FinBank)
	T r a n s f o r m i n g	Reconfiguring IR procedures	"Being able to measure the cybersecurity operations or processes using key risk indicators is crucial. So, until we get to a point where we have meaningful key risk indicators that are measurable, comparable, informational, and predictable, we cannot get full benefit out of analytics." (Manager of Cybersecurity Strategy and Governance, FinInsuranceA)
		Redesigning IR workflows	"So, by virtue of our teams periodically reviewing and measuring our capabilities against the framework and against our industry peers, we are demonstrating that we are proactively trying to continue to improve our capabilities to stay one step ahead of the threats." (Senior IT Risk Partner, FinInsuranceA)
		Improving IR maturity	"The first step we did to improve the analytical capability in incident response process was to collaborate and establish a relationship with our analytics team. We started embedding some of their practices and processes and even using some of their services. By building stronger relationships and leveraging the capabilities of analytics team, we were able to improve and drive our decisions based on the data." (Director of Global Cyber Forensics, FinInsuranceB)

Aggregate dimension	Second-order themes	Representative quotes
<i>Agile IR strategies</i>	Active threat reconnaissance	"Using [cyber] threat intelligence, our incident response team tries to understand the methodology, intent and focus of the attack and in some cases, can detect an incident during the reconnaissance phase." (General Manager of Cybersecurity Strategy and Governance, FinInsuranceA)
	Active threat defence	"Advanced attackers are continuously innovating and so must we. They are using artificial intelligence and machine learning to carry out the targeted attacks...not using these methods in our cyber defence potentially puts our organisation behind the attackers." (General Manager of Cybersecurity Risk Management, FinBank)
	Pervasive learning	"We need clear visibility into data regarding system configuration and patch levels, device behavior, vulnerabilities, and overall cybersecurity state. The information regarding the cybersecurity state and vulnerabilities for these systems needs to be continuously monitored and correlated to demonstrate security compliance when required." (Cybersecurity Senior Manager, FinBank)
<i>Evolving cyber threat environment</i>	Complex and dynamic threats	"Sophisticated, well-resourced advanced persistent threat actors are impossible to stop from intruding. Having advanced analytics capability means that we can capture and use far more data than we could before and therefore understand these types of threats in a better way." (Chief Information Security Officer, FinBank) "Attackers use complex and dynamic ways to break into our networks. For example, they will use zero-day attacks, stolen access credentials, infected mobile devices, a vulnerable business partner, or other tactics to carry out the cyber-attack. Having a real-time analytics capability means that we have the ability to detect such attacks in real-time and respond to them in a proactive way". (Security Operations Centre Manager, FinBank)

Appendix C: Agile cybersecurity incident response in action: an illustrative example

The following example described by FinBank's Chief Information Security Officer illustrates how their IR team detected and discerned diverse types of phishing attacks and responded to them in an agile manner using RTA-based sensing, seizing, and transforming dynamic IR capabilities:

Phishing and spear-phishing attacks are the most common attacks we encounter on [a] daily basis ... But spear-phishing attacks, in particular, are highly targeted and can be used to begin an APT attack as these are highly personalised emails and are very challenging to detect because they appear to come from sources close to the target.

Therefore, FinBank needed to differentiate traditional phishing attacks, which were impersonal and sent in bulk, from spear-phishing attacks, which were highly personalised, targeting high-level individuals or groups within their organisation. This participant further explained:

Our tier-1 security analysts are using continuous monitoring systems (SIEM) to monitor our communication servers and also to analyse events in which we do event and log collection and correlation. The aim is to detect any potential phishing emails circulating in our network. (FinBank, Chief Information Security Officer) *[Sensing]*

Once an attack was detected by tier-1 analysts, tier-2 analysts used threat hunting to understand the attack in a broader context and determine the type of phishing (spear or traditional). They also used the kill chain to classify the events and activities and prepared the incident investigation and risk assessment reports. *[Seizing]*

Tier-3 security experts then reviewed the incident discovery and risk assessment reports and executed the most suitable response (active threat defence in this case) as an agile response strategy and improved the incident management processes, workflows, and procedures based on the lessons learnt. *[Transforming]*

FinBank's Chief Information Security Officer concluded this example as follows:

Real-time threat intelligence and situation awareness in this case acted like an early warning system. It helped us to identify what the next credible or significant threat to us might be. We then took [an] agile approach and implemented additional controls to prevent them from happening. So, it is managing the risk in a proactive way so that it does not become an issue.

Table 3 provides a summary of the main trigger, the dynamic IR capabilities, and the outcome discussed in the aforementioned example.

Table 3. Agile incident response in action

Main trigger	Dynamic IR capabilities			Outcome
	Sensing	Seizing	Transforming	
Deciphering phishing and spear-phishing attacks	• Threat intelligence • Continuous monitoring	• Dynamic risk assessment • Threat hunting	• Reconfiguring IR procedures • Redesigning IR workflows	Created innovative, swift, and flexible incident management workflows, processes, and procedures to manage phishing and spear-phishing attacks

Thus, our findings highlight that RTA-based sensing, seizing, and transforming dynamic IR capabilities, together as a system of capabilities, played a critical role in improving the agility of the IR process. This, in turn, helped focal organisations to respond to the dynamic cyber threat environment in a swift, flexible, innovative, effective, and proactive manner.